RIC JOSE FRANCIS MOSCOSO FELIPE

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CAREER SUMMARY

Electrical and Electronics Engineering graduate with hands-on experience in test engineering, PCB troubleshooting, and electronics manufacturing within a high-mix production environment. Proven ability to support ICT, functional testing, and NPI processes, with a strong focus on failure analysis, system validation, and production efficiency. Seeking to contribute to test engineering or electronics manufacturing roles in Singapore and Malaysia.

SKILLS

Programming: Object-Oriented Programming, C++, LTSPICE, Arduino (Embedded C), MATLAB
Hardware & Design: PCB Design (KiCAD, EasyEda) AutoCAD, Soldering, Circuit Debugging
Testing and Manufacturing: ICT (Keysight i3070), AOI, MDA (TRI5001E), GOPEL Juliet, Functional Testing
Concepts: Troubleshooting, Root Cause Analysis, NPI Support
Languages: English (Fluent), Malay Language and Filipino (Native)

WORK EXPERIENCE

PLEXUS MANUFACTURING SDN. BHD. (HILLSIDE) Dec 2024 – April 2025
Test Engineer Intern

- Validated Functional Testing using 8 gold units during NPI phase, ensuring test accuracy, repeatability, and compliance with medical manufacturing standards prior to mass production
- Performed functional testing for New Product Introduction (NPI) units, identifying early-stage defects and reducing downstream production risks
- Configured and optimized In-Circuit Test (ICT) and Functional Test systems, enabling stable and efficient production testing
- Collaborated with engineers and technicians to troubleshoot test failures and enhance system reliability

PLEXUS MANUFACTURING SDN. BHD. (SEASIDE) Dec 2022 – April 2023
Test Engineer Intern

- Diagnosed and resolved 7+ PCB failures per day, improving troubleshooting turnaround time and maintaining production flow
- Set up and calibrated 2 ICT systems daily, enhancing test reliability and accuracy in PCB validation
- Reduced test downtime by identifying and resolving tester-related issues, supporting continuous manufacturing operations
- Conducted visual inspections and debugging of PCBAs to identify soldering and component defects

PROJECTS

AI-Based Storage Tracking System Using R-CNN and RFID Integration in collaboration with Analog Devices Inc.: April 2025 – Dec 2025

- Developed an R-CNN-based object detection system for warehouse inventory tracking
- Enhanced classification accuracy through iterative model tuning under varying environmental conditions
- Integrated RFID technology for real-time tracking, enhancing visibility in Industrial IoT applications
- Collaborated with industry engineers to align system design with scalable deployment standards

Motor Directional Control using Plug and Play Relay Control — Aug 2023 – Dec 2023

- Designed and implemented a bidirectional motor control circuit using a Plug-and-Play Relay configuration to execute forward, reverse, and braking functions
- Improved a "Plug and Play" interface that allows for rapid component replacement and modular system expansion to minimize maintenance downtime
- Applied opto-isolation techniques to protect control circuits from high-voltage noise and electrical surges

Low Cost IOT-Based Home Temperature Monitoring System — Aug 2023 – Dec 2023

- Built a real-time temperature and humidity monitoring system using ESP32 and DHT22 sensors
- Developed embedded firmware in C++ for data acquisition, filtering, and automated control responses
- Enabled remote monitoring via Blynk platform, improving accessibility and system usability.

EDUCATION

INTI INTERNATIONAL COLLEGE PENANG

Bachelor of Engineering (Hons) in Electrical & Electronic Engineering 3+0 in collaboration with Coventry University, UK — Jan 2024 – Dec 2025

- Upper Second Class Honours (CGPA:3.33)

Diploma in Electrical and Electronic Engineering — Aug 2021 – Dec 2023

CO-CURRICULUM

INTI INTERNATIONAL COLLEGE PENANG

Vice-President of INTI Engineering Club (IEC) — Jan 2024 – Jan 2025

- Led engineering initiatives and coordinated student engagement activities

ACHIEVEMENTS & COMPETITIONS

- 1st Place – Integrated Design Project Short Video Competition 2024
- Participant – Air Selangor Data & Digital Hackathon 2024
- Participant – International STEM Innovation Competition 2024
- Participant – Young Innovation Challenge 2024
- Participant – Shell Selamat Sampai Varsity Challenge 2024

CERTIFICATIONS & DEVELOPMENT

- INTEL University VLSI Bootcamp Series Post-Silicon Validation 2024

OTHERS

- Achieved Black Belt Shodan (1st Dan) in Karate Club
- Strong interest in semiconductor, electronics manufacturing, and test engineering environments